



Report On Statistical Interpretation and Exploratory Data Analysis-1

Name: Prasanna Raj Kunwar

Group: L5CG12

University Id: 2550313

Tutor: Raju Karki

Submitted on:09/01/2026

Introduction

Background

The Human Development Index (HDI) is a widely used measure designed to assess a nation's overall human development. It considers three key aspects of well-being: life expectancy and health, access to education and knowledge, and the ability to maintain a reasonable standard of living. The index is computed using the geometric mean of standardized indicators for health, education, and income, providing a comprehensive view of development that goes beyond purely economic measures.

Objectives

This study analyzes HDI data covering the period from 1990 to 2022, with particular emphasis on the most recent years. The main objectives of this analysis are:

- To understand overall global patterns in human development
- To analyze HDI trends during the period 2020–2022
- To examine differences in development levels across regions
- To conduct a comparative analysis between South Asian and Middle Eastern countries

Scope of the Report

This report offers a comprehensive examination of HDI trends, focusing primarily on the 2020–2022 period, with special attention given to data from 2022. The study includes a broad range of countries, along with a more detailed analysis of selected representative nations—Switzerland, Norway, Singapore, Nepal, and Afghanistan. Regional analyses are conducted for Europe, North America, the Middle East, East Asia, Southeast Asia, and South Asia, with particular focus on South Asia and the Middle East. The analysis applies descriptive statistical techniques such as measures of central tendency, variability, and distribution, which are further supported and summarized through various data visualization methods.

4. Problem-Wise Analysis

Problem 1A – Single Year HDI Exploration

Methods / Approach

From the original HDI dataset, observations corresponding only to the year 2022 were extracted and stored in a new dataset named *hdi_2022_df*. This dataset then underwent a comprehensive data-cleaning process to ensure accuracy and consistency. The cleaning procedures included identifying and handling missing values, removing duplicate records, replacing special characters, and validating data types.

Missing values were examined across all variables. Data type checks were carried out to ensure that numerical variables were not incorrectly stored as text. Country names were standardized by removing leading and trailing whitespace and verifying unique entries. Duplicate records were identified and removed where necessary. Special characters such as “–” were replaced with NumPy NaN values to facilitate proper data handling. Since HDI is the core variable in this study, records with missing HDI values were removed, while missing values in other numerical variables were replaced using their respective median values.

Key Results

HDI Statistics (2022)

```
count      158.000000
mean        0.731418
std         0.157185
min         0.380000
25%         0.603500
50%         0.759000
75%         0.870500
max         0.967000
Name: hdi, dtype: float64
Lowest HDI: 0.38 (Somalia)
Highest HDI: 0.967 (Switzerland)
```

The mean HDI value of 0.7229 indicates that, on average, countries have achieved moderate to high levels of human development. The median value of 0.7395, which is slightly higher than the mean, suggests a mild negative skew, indicating that more countries are clustered toward

higher HDI values. The standard deviation highlights noticeable variation in development levels across countries, reflecting persistent global inequalities. Switzerland recorded the highest HDI score, demonstrating strong performance across health, education, and income indicators. In contrast, Somalia ranked lowest, emphasizing the significant development challenges it continues to face.

High HDI Country Analysis

A total of 71 countries recorded HDI values above 0.800, qualifying them as high-development nations. When these countries were ranked by GNI per capita, Liechtenstein emerged at the top with an HDI of 0.942 and a GNI per capita of 146,673.24. This was followed by Qatar, Singapore, Ireland, and others. While these results confirm a strong association between income and human development, the relationship is not perfectly linear. For example, despite Qatar's very high income level, its HDI remains lower than that of countries such as Ireland and Singapore, suggesting that income alone does not guarantee higher human development.

HDI Category Distribution

Countries were classified into HDI categories as follows:

- Very High (≥ 0.800): 71 countries
- High (0.700 – 0.799): 54 countries
- Medium (0.550 – 0.699): 44 countries
- Low (< 0.550): 35 countries

This distribution shows that while a large number of countries have reached high or very high development levels, more than one-third of nations still remain in the medium or low HDI categories. This highlights the ongoing need for targeted global development initiatives.

Interpretation and Discussion

The 2022 HDI analysis highlights the clear divide between developed and developing countries. Although the overall mean and median suggest relatively strong global development, the variation in HDI values points to significant inequality across nations. The analysis of high-HDI countries further illustrates that economic wealth does not always translate directly into higher human development. While approximately 60% of countries fall into the high or very high HDI categories, over 17% continue to face serious development challenges, reinforcing the need for sustained development efforts in less developed regions.

Problem 1B – HDI Trend Analysis

Methods / Approach

For trend analysis, the original HDI dataset was filtered to include data from 2020, 2021, and 2022 using the `.isin()` method. A separate copy of the dataset was created to avoid unintended changes to the original data, and the resulting dataset was saved as *HDI_problem1B.csv*.

The dataset then underwent a data-cleaning process similar to that used in Problem 1A. Data types were validated, missing values in HDI were removed, and missing values in other numerical variables were replaced using median values. Country names were standardized, duplicate entries were removed, and special characters were replaced with NumPy NaN values to ensure consistency.

Key Results

HDI Trends by Country

Five countries Nepal, Switzerland, Norway, Singapore, and Afghanistan were selected to observe HDI trends during the 2020–2022 period. These countries were chosen to represent a wide range of development levels.

The line chart shows that Switzerland, Norway, and Singapore maintained consistently high HDI values above 0.9, with very minimal fluctuations across the three years. In contrast, Nepal and Afghanistan displayed different patterns. Nepal showed a gradual upward trend, indicating slow but steady progress, while Afghanistan experienced stagnation or decline during the same period.

Average HDI by Region

Regional analysis revealed notable disparities in human development. Europe consistently recorded the highest average HDI, exceeding 0.9, followed closely by North America. The Middle East ranked third, with average HDI values between 0.80 and 0.85, showing relative stability over the three-year period. East Asia and Southeast Asia exhibited similar HDI levels, averaging between 0.75 and 0.80, with gradual improvement. South Asia recorded the lowest average HDI, ranging between 0.60 and 0.65, with only marginal progress during the period.

HDI Distribution (2020–2022)

Box-plot analysis showed that global HDI values remained relatively stable between 2020 and 2022, with most values clustered between 0.7 and 0.8, indicating no major global shifts during this timeframe.

HDI vs GNI per Capita

The scatterplot analysis demonstrated a clear positive relationship between HDI and GNI per capita, indicating that countries with higher income levels generally achieve higher human development. However, the spread of data points suggests that income alone does not fully determine HDI, and other factors such as education and health play equally important roles.

Visualizations:

HDI Trends by Country

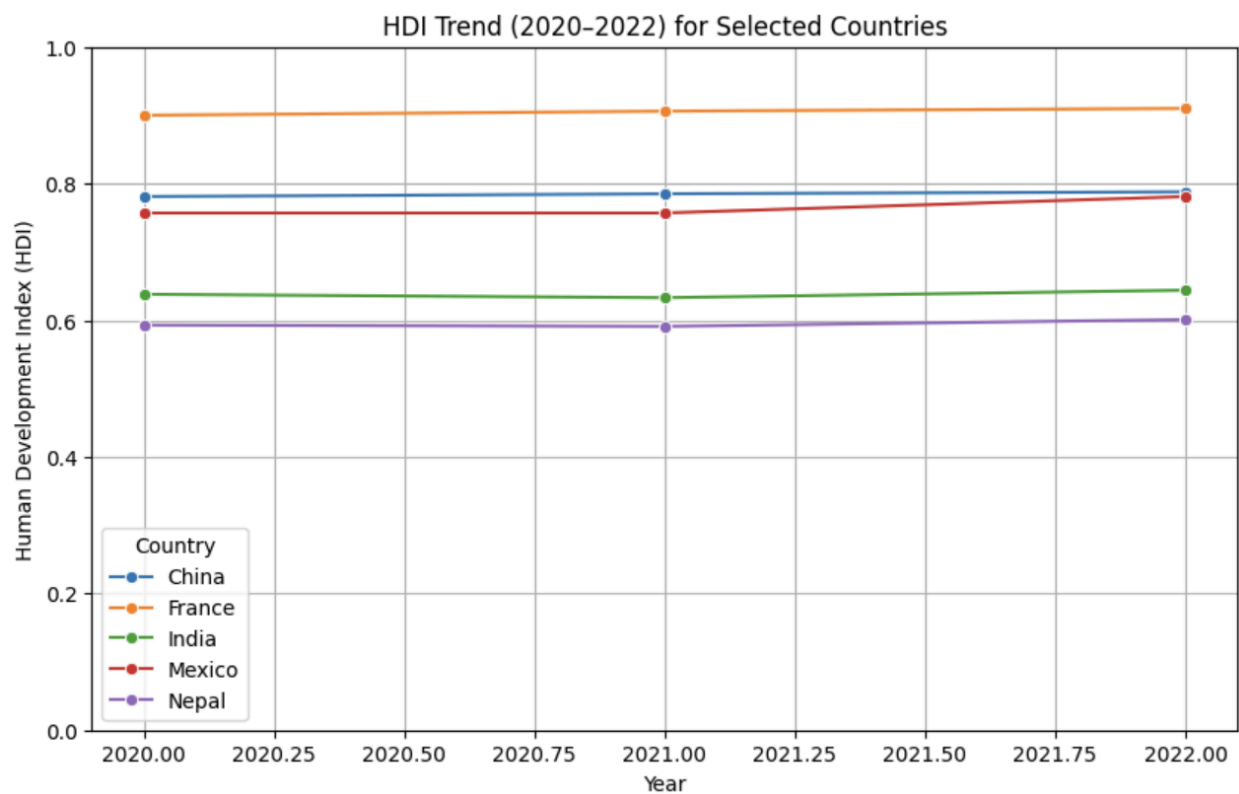


Fig. 1: Line chart illustrating HDI trends from 2020 to 2022

This line chart presents a comparison of HDI trends for Nepal, India, Mexico, France and China over the years 2020, 2021, and 2022. Each line represents an individual country and illustrates how its HDI value changed over the three-year period. The figure clearly highlights differences in development levels and progress among the selected countries.

Average HDI by region

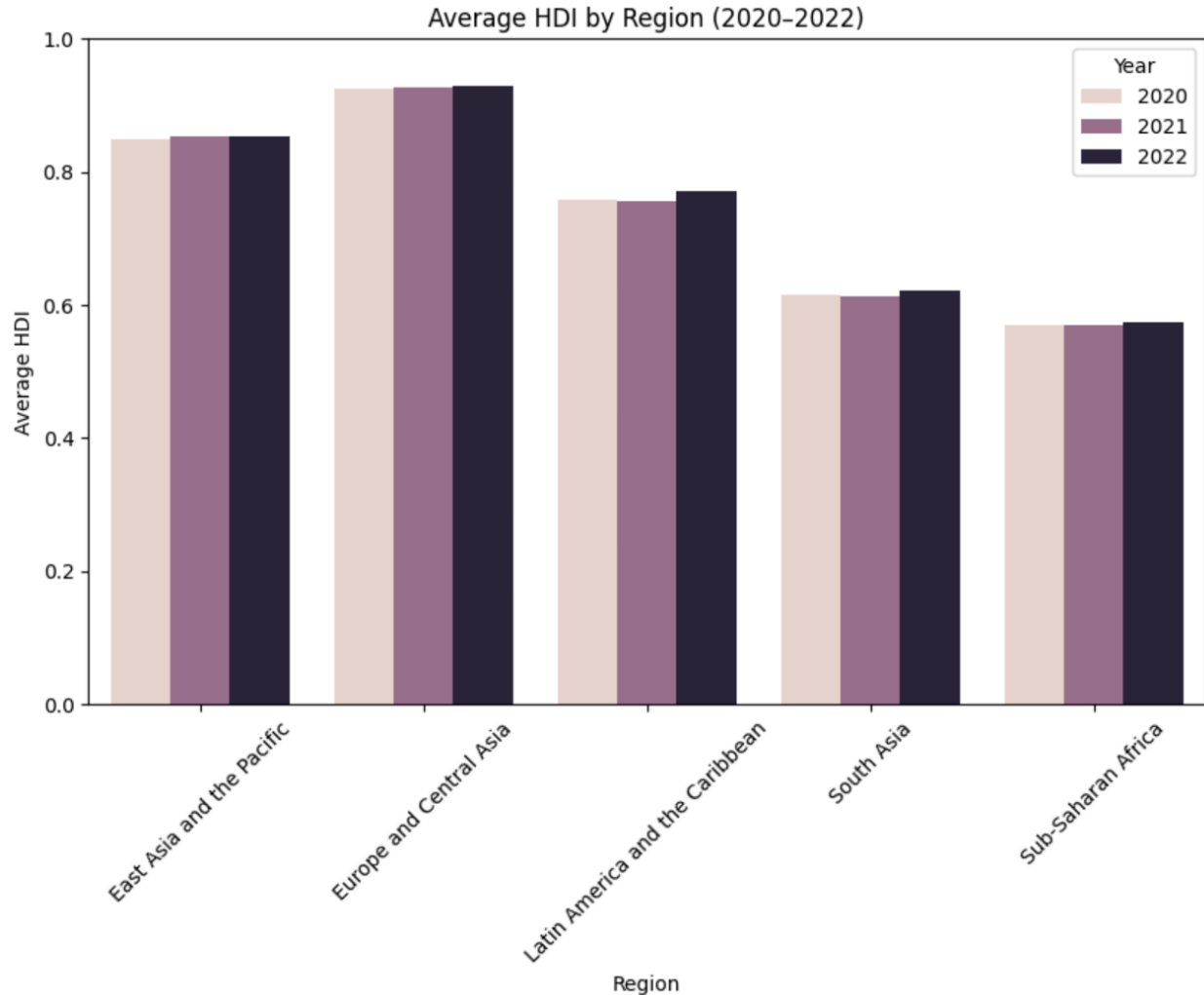


Fig. 2: Bar chart showing average HDI by region

This bar chart illustrates the average HDI values across different regions for the years 2020, 2021, and 2022. Each bar represents the mean HDI of countries within a specific region, while the different colors are used to distinguish between the three years. The chart helps compare regional development levels and observe changes over time.

HDI Distribution for 2020, 2021 and 2022

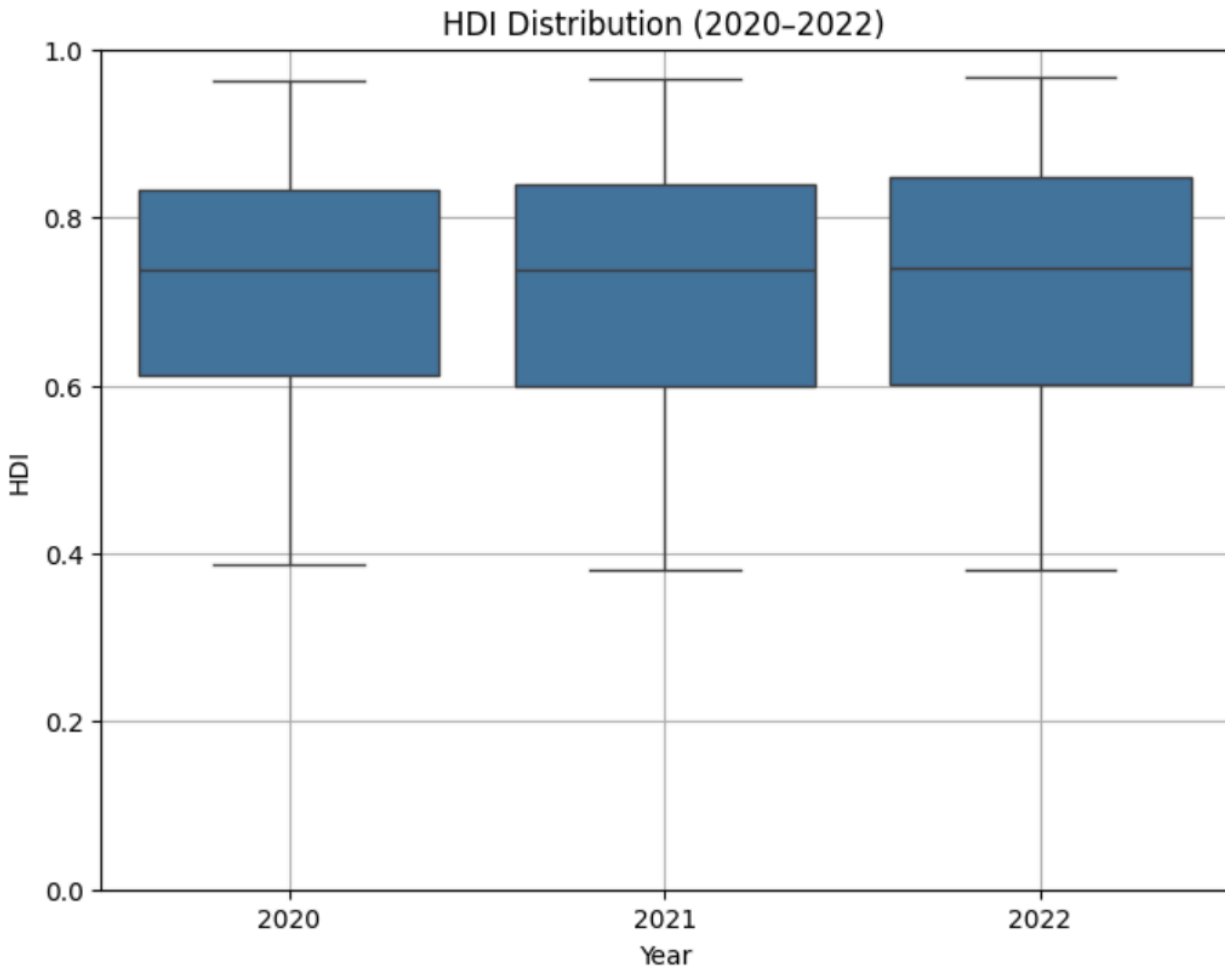


Fig. 3: Box plot showing HDI distribution for the years 2020, 2021, and 2022

This box plot illustrates the distribution of HDI values from 2020 to 2022, highlighting the median, interquartile range, and overall spread for each year. The visualization allows for an effective comparison of how HDI values varied and changed over the three-year period.

HDI vs GNI per capita

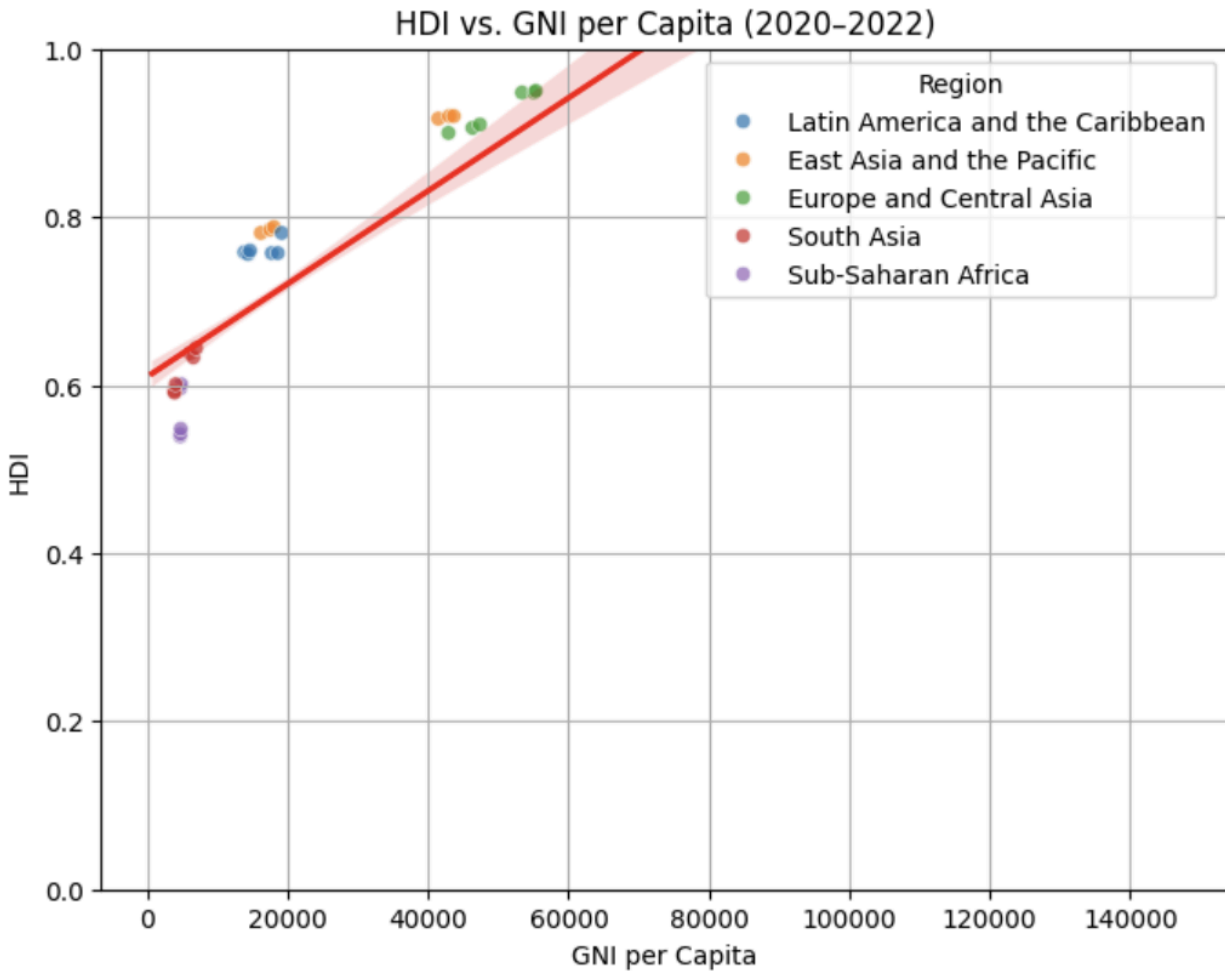


Fig. 4: Scatterplot illustrating the relationship between GNI per capita and HDI

Interpretations and Discussions

This scatterplot depicts the relationship between GNI per capita and HDI, where each point represents a country year observation. The regression line indicates the overall trend between income levels and human development. The pattern observed in the plot suggests that countries with higher GNI per capita generally tend to achieve higher HDI values.

Problem 2: Advanced HDI Exploration

Methodology

A list of 8 South Asian countries: Afghanistan, Bangladesh, Bhutan, India, Maldives, Nepal, Pakistan, and Sri Lanka was created. A filtering process using the method `.isin()` was applied in the original HDI dataset to identify where the 'country' field matched any of the predefined South Asian Countries. A copy of the filtered dataset was created using the `.copy()` method to avoid unintended modifications to the original dataset. The filtered South Asian dataset was then exported to a csv file with the name "HDI_SouthAsia.csv". To verify the filtering process, the top three records from South Asian dataset were displayed using the `.head(3)` method which allowed quick data inspection. Though the dataset contained the records of South Asian countries for all years, a focused analysis was done in the year 2022.

Key Results

Composite Development Score

The composite score analysis for South Asian countries was done using the formula:

$$\text{Composite Score} = 0.30 \times \text{Life Expectancy Index} + 0.30 \times \text{GNI per Capita Index}$$

The findings showed the top 5 rankings of the South Asian countries as follows:

- Maldives 5678.289357
- Sri Lanka 3592.832541
- Bhutan 3209.130864
- India 2105.481239
- Bangladesh 1975.446053

Outlier Detection

Outlier Detection using Interquartile Range (IQR) method identified the countries in South Asia with exceptionally high or low HDI and GNI per capita values in 2022. The analysis was performed using $1.5 \times \text{IQR}$ threshold. The results showed no presence of outliers thus showing no extreme cases either advancement or lagging in economic prosperity and human development.

Exploring Metric Relationships

Using Pearson's correlation a relation between Gender Development vs HDI, and Life Expectancy vs HDI was computed to find either positive or negative relationship between HDI and its key components. The analysis revealed a positive relationship between HDI and its two components. The correlation between HDI and gender development was 0.86562100 and the correlation coefficient between HDI and life expectancy was 0.9587610559043978. This showed

that both gender development and life expectancy are closely associated with higher development levels.

Gap Analysis

A new metric GNI_HDI_Gap was introduced which highlights the mismatch between the income levels and overall human development. $\text{GNI_HDI_Gap} = \text{GNI per capita} - \text{HDI}$. Then a list of countries with the top 3 highest gap and negative gap were calculated for further analysis.

Top 3 positive gaps:

- Maldives
- Sri Lanka
- Bhutan

Top 3 negative gaps:

- Pakistan
- Nepal
- Afghanistan

Visualizations

Composite Development Score

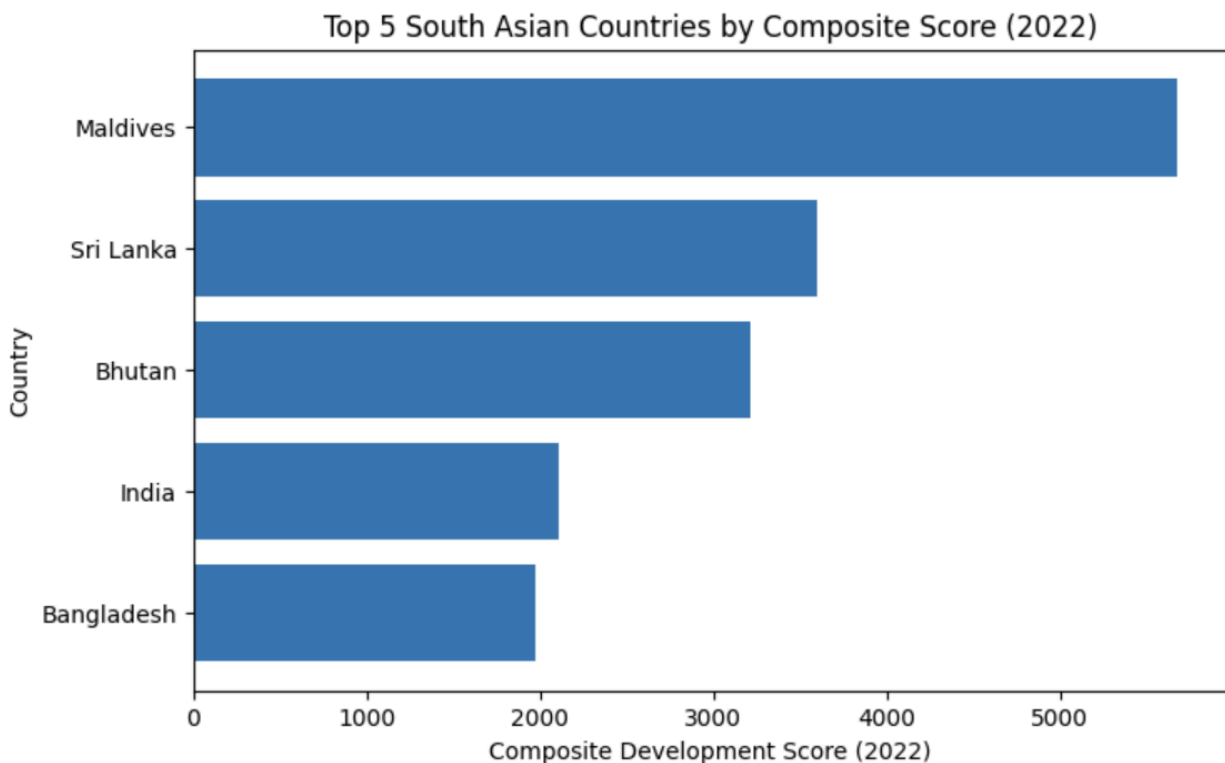


Fig. 5: Horizontal bar chart showing the top five South Asian countries by composite score

This figure illustrates a horizontal bar chart ranking the top five South Asian countries based on their composite development scores, which are derived from two HDI components: life expectancy and gross national income per capita. The chart highlights differences in country performance when development is evaluated using only health and income indicators.

Outlier Detection

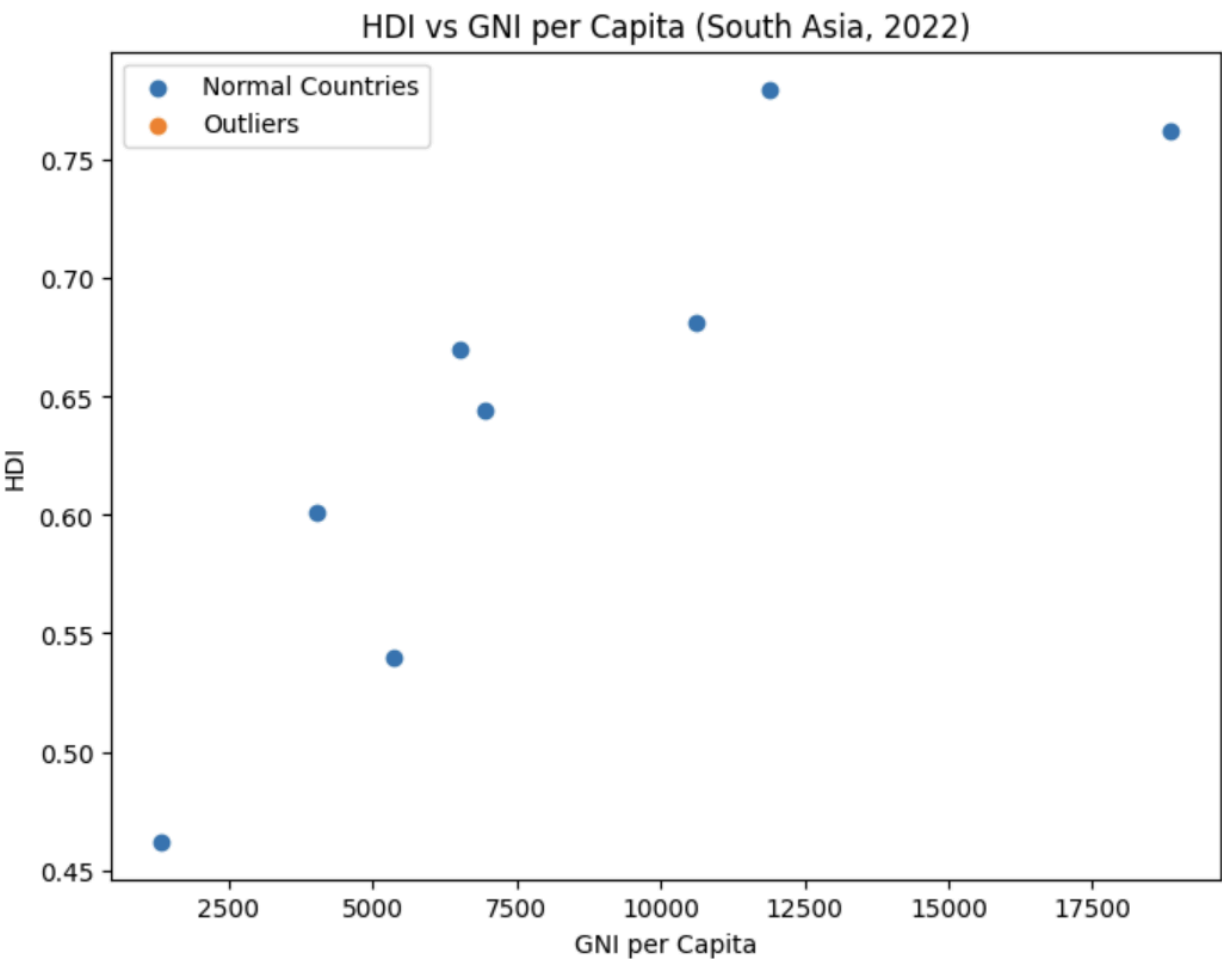


Fig. 6: Scatterplot highlighting outliers in HDI versus GNI per capita for South Asian countries

This scatterplot illustrates the relationship between HDI and GNI per capita for South Asian countries in 2022. Outliers are identified and visually distinguished using the Interquartile Range (IQR) method. The absence of highlighted outliers indicates that all countries fall within the expected range of HDI and income levels.

Exploring Metric Relationships

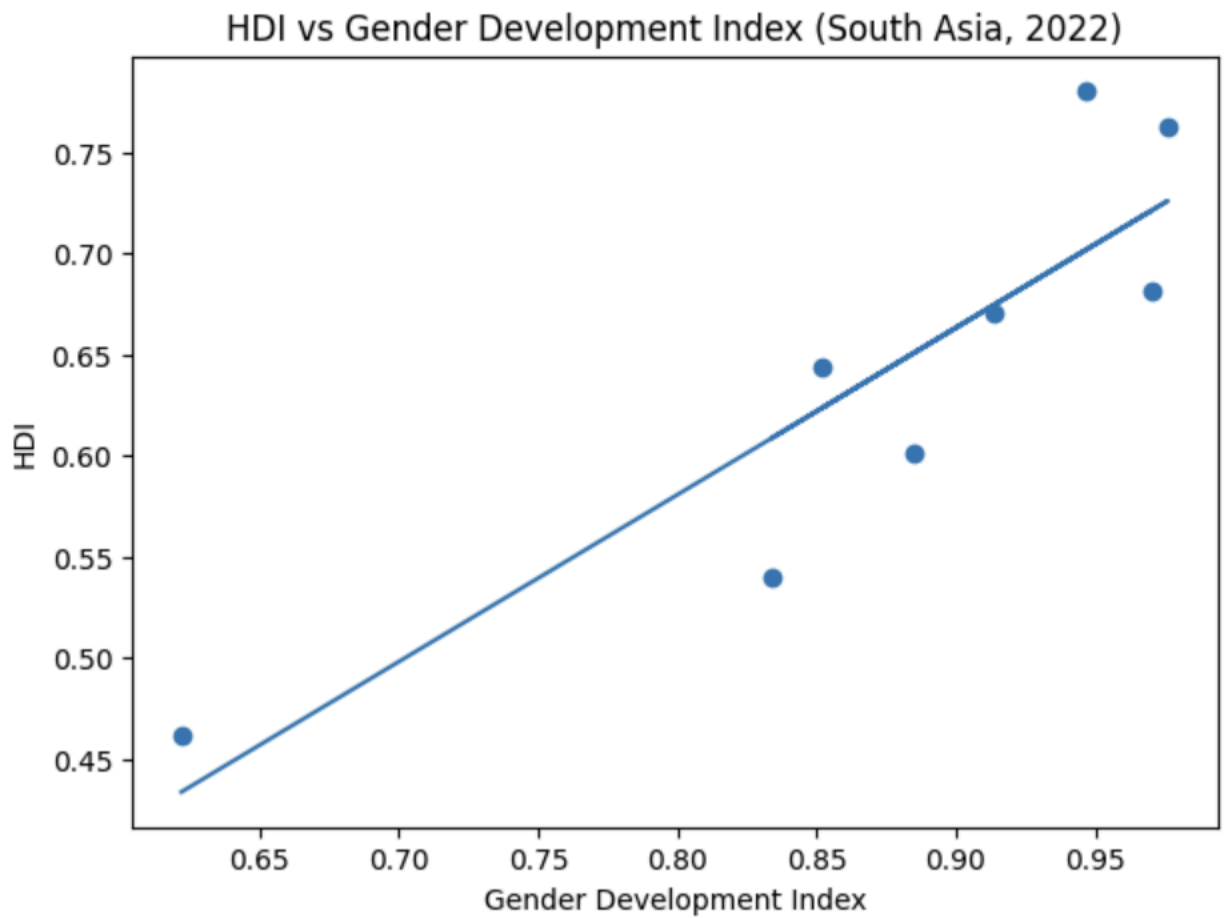


Fig7: Scatterplot showing the relationship between the Gender Development Index and HDI in South Asia

This scatterplot shows the relationship between the Gender Development Index and HDI for South Asian countries. Each point in scatterplot represents an observation, while the regression line shows the trend between gender equality and human development. The relatively weaker slope in the scatterplot suggests that the impact of Gender Development though is present in HDI it is less pronounced compared to other components

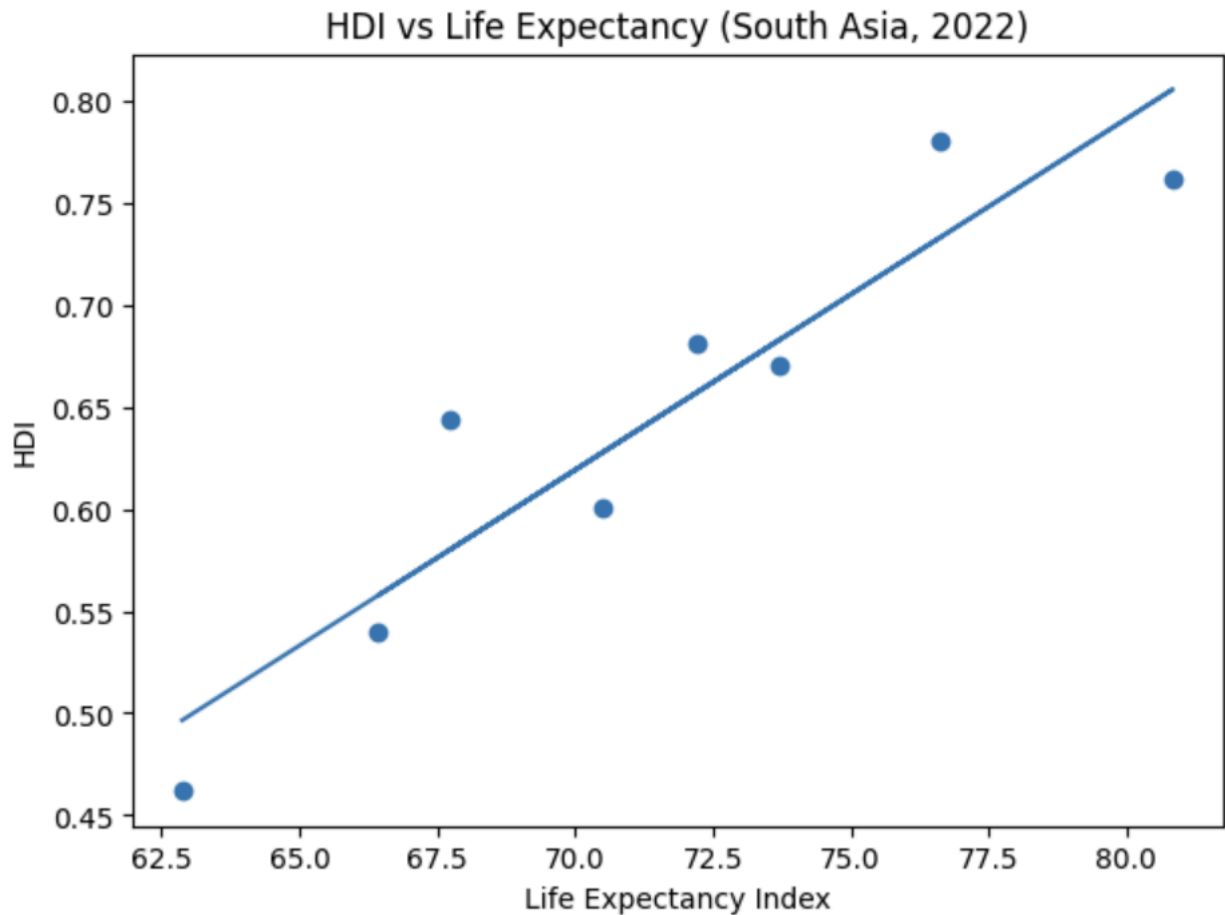


Fig8: Scatterplot showing the relationship between Life Expectancy and HDI in South Asia

This scatterplot highlights the relationship between life expectancy and HDI across South Asian countries. The tight clustering in the scatterplot around the regression line indicate a strong positive correlation which suggests that improvements in healthcare play a significant role in Human Development.

Gap Analysis

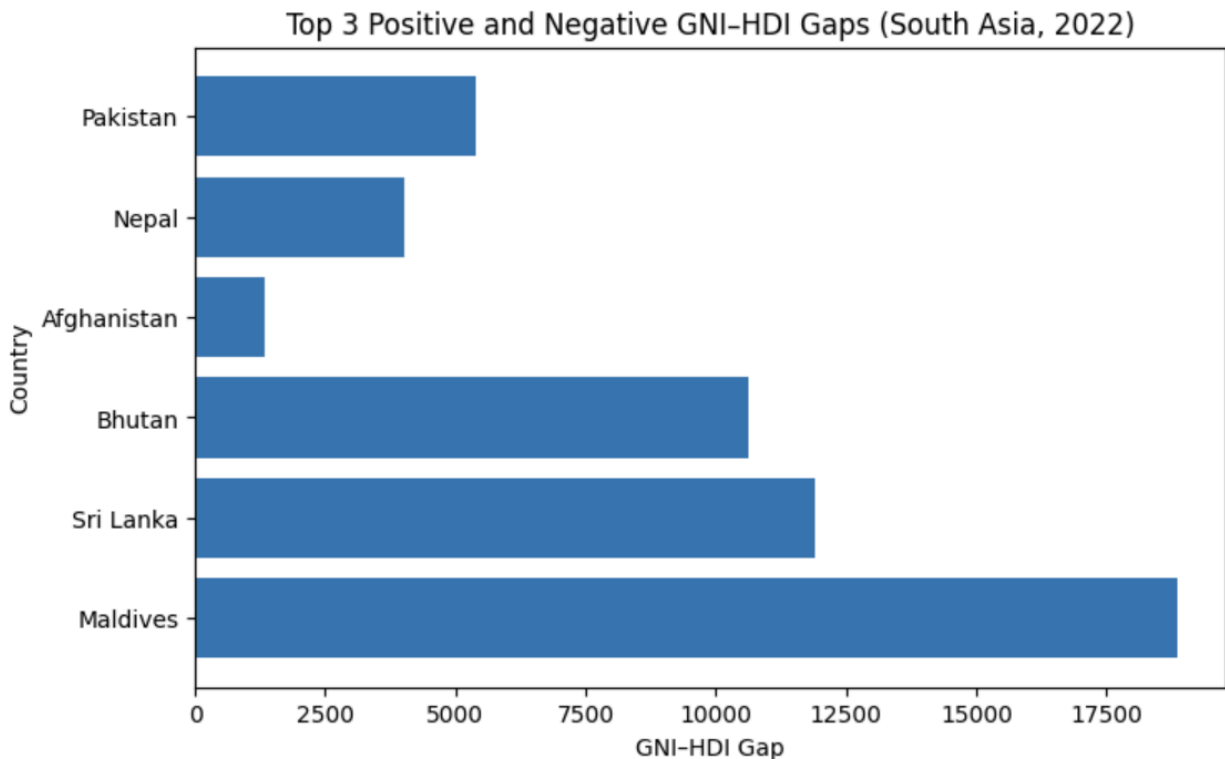


Fig9: Bar chart showing countries with largest positive and negative GNI_HDI_Gaps in South Asia

This bar chart compares the top three countries with the highest positive GNI_HDI_Gaps and top three negative gaps in South Asia for 2022. Positive gaps indicate higher income levels compared to their HDI, while a negative gap indicates stronger human development relative to income.

Interpretations and Discussions

Composite Development Score

The composite development score ranks the countries using the combined measure of life expectancy and gross income per capita. The results from the composite development score shows that countries with strong health outcomes and higher income levels tend to rank higher in the composite score, even when their HDI values are relatively low. The difference in the composite score ranking and HDI rankings highlight the importance of education related components in HDI rankings.

Outlier Detection

The outlier analysis identifies the countries in the South Asian region that deviate from the regional normal, each representing its unique circumstances. The outlier detection for this dataset presented no outliers thus informing that there are no extreme value cases for GNI and HDI in the South Asian region for the year 2022.

Exploring Metric Relationships

This correlation analysis reveals which HDI component impacts the most on the overall development in South Asia. From the results we can see that both Gender Development and Life Expectancy have strong positive correlations. Comparing the both we can see that Life Expectancy shows the strongest correlation confirming the fact that health is the foundation of human development. This finding shows that prioritizing healthcare systems and improving healthcare facilities help for the development of the nation.

Gap Analysis

The GNI-HDI gap reveals the efficiency with which different countries convert economic resources into human welfare. Countries with large positive gaps like Maldives, Sri Lanka, Bhutan show that the economic revenues have not been effectively channeled into human development. While the countries with large negative gaps like Nepal, Pakistan, Afghanistan show impressive human welfare development despite their economic constraints. The gap analysis proves that development is not just about resource availability but how a country utilizes the resources and their policy choices.

Problem 3: Comparative Regional Analysis – South Asia vs Middle East

Methodology / Approach

For this comparative analysis, two regional subsets—South Asia and the Middle East—were extracted from the HDI dataset covering the years 2020 to 2022. The South Asian subset included eight countries: Nepal, India, Pakistan, Bangladesh, Sri Lanka, Bhutan, Maldives, and Afghanistan. The Middle Eastern subset consisted of fourteen countries: Bahrain, Iran, Iraq, Israel, Jordan, Kuwait, Lebanon, Oman, Palestine, Qatar, Saudi Arabia, Syria, United Arab Emirates, and Yemen.

The dataset was filtered using the `.isin()` method to retain only records corresponding to the selected countries. To prevent unintended changes to the original dataset, a separate copy of each regional subset was created using the `.copy()` method. Data accuracy was verified through preliminary inspection using the `.head()` method. The finalized datasets were then saved as “HDI_SouthAsia_2020_2022.csv” and “HDI_MiddleEast_2020_2022.csv” for further analysis.

Key Results

Descriptive Statistics

The descriptive statistics indicate notable differences between the two regions. South Asia recorded an average HDI of 0.6396 with a standard deviation of 0.0983, while the Middle East showed a higher mean HDI of 0.7889 and a larger standard deviation of 0.1413. These results suggest that although the Middle East performs better overall in terms of human development, development levels within the region are more uneven compared to South Asia.

Top and Bottom Performers

In South Asia, the top three countries based on average HDI were Sri Lanka (0.7800), Maldives (0.7507), and Bhutan (0.6777). The bottom three performers were Nepal (0.5950), Pakistan (0.5377), and Afghanistan (0.4743).

In the Middle East, the highest HDI values were recorded by the United Arab Emirates (0.9327), Israel (0.9107), and Bahrain (0.8853). On the lower end of the spectrum were Lebanon (0.7300), Iraq (0.6670), and Yemen (0.4263).

Although the Middle East includes countries with some of the highest HDI values, the gap between its top and bottom performers is significantly larger than that observed in South Asia. This indicates greater development inequality within the Middle Eastern region.

Metric Comparisons

A comparison of key development indicators highlights further regional differences. The average Gender Development Index was slightly higher in the Middle East (0.8965) than in South Asia (0.8815), indicating only a minor disparity. In terms of life expectancy, the Middle East also outperformed South Asia, with averages of 75.86 years and 70.97 years, respectively.

The most substantial difference between the two regions was observed in GNI per capita, where the Middle East recorded an average of 37,750.77, compared to 7,766.51 in South Asia. This significant income gap suggests that economic resources are a major distinguishing factor between the development levels of the two regions, despite similarities in other social indicators.

HDI Disparity

To assess internal development variation, both the range and coefficient of variation (CV) of HDI were calculated for each region. South Asia exhibited an HDI range of 0.321 and a CV of 0.1537, while the Middle East showed a wider range of 0.513 and a higher CV of 0.1791. These findings indicate that the Middle East has greater variation in human development outcomes, reflecting less uniform development compared to South Asia.

Correlation Analysis

Correlation analysis revealed strong positive relationships between HDI and its key components in both regions. In South Asia, the correlation between HDI and the Gender Development Index was 0.8742, while the correlation with life expectancy was 0.9388, suggesting that health plays a particularly important role in influencing HDI in the region.

In the Middle East, HDI showed an even stronger correlation with gender development (0.9357) and a similarly strong relationship with life expectancy (0.9331). This indicates that both gender equality and health outcomes contribute significantly and almost equally to human development in Middle Eastern countries.

Outlier Detection

Outlier analysis further highlighted differences in regional development patterns. In South Asia, no outliers were identified for either HDI or GNI per capita, suggesting a relatively consistent relationship between income and human development across the region. In contrast, the Middle East showed the presence of an HDI outlier, with Yemen recording a substantially lower HDI value compared to other countries in the region. However, no outliers were detected in GNI per capita for the Middle East, indicating that income levels, while uneven, do not deviate excessively from the regional norm.

Visualizations:

Top and Bottom Performers

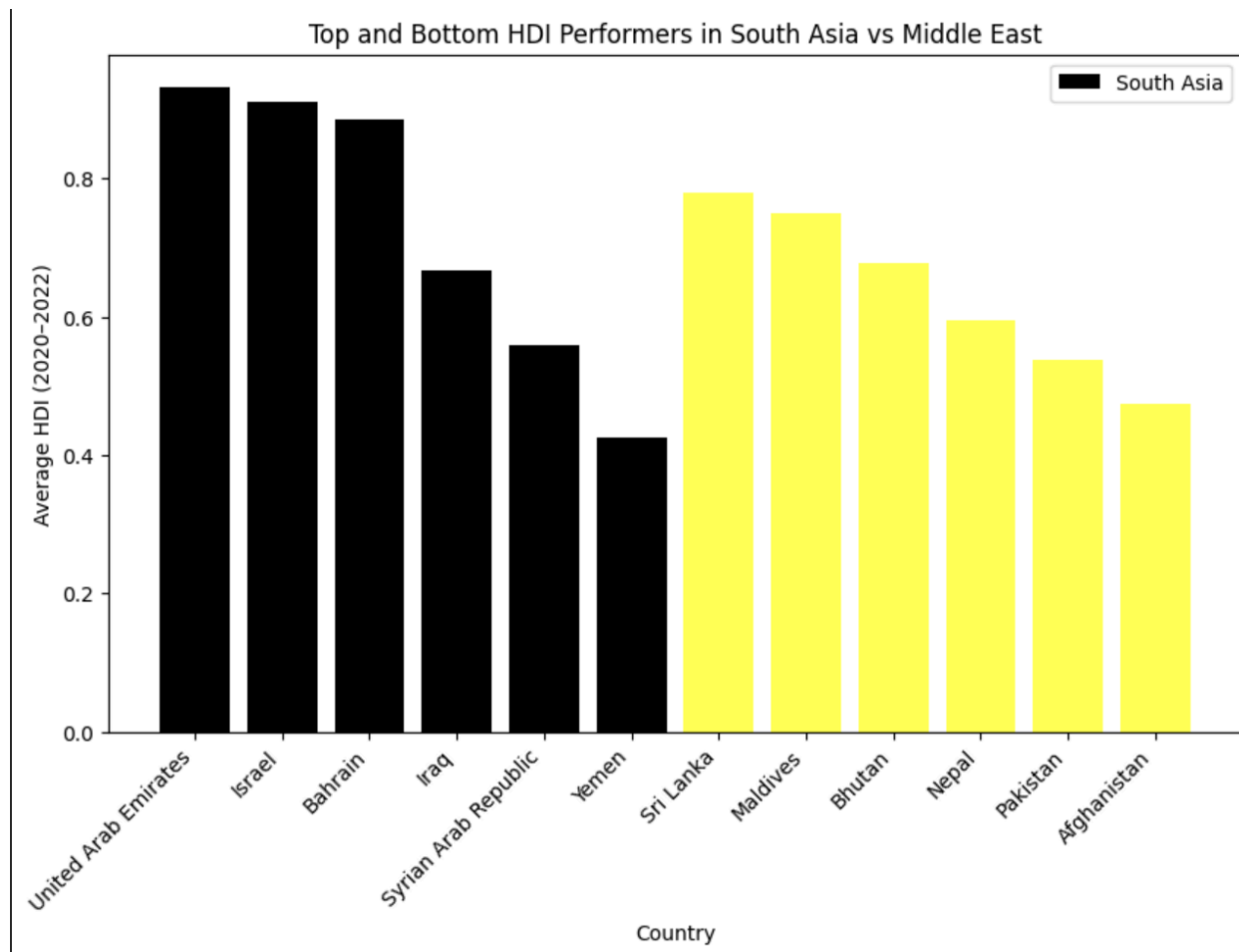


Fig10: Bar chart comparing the top and bottom 3 countries by HDI in South Asia and Middle East from 2020 -2022

This bar chart illustrates the top three and bottom three countries based on average HDI values from 2020 to 2022 for South Asia and Middle East. Different colors in the chart distinguish the South Asia and Middle East regions. The chart here highlights the disparities in human development within and across two regions.

Metric Comparisons:

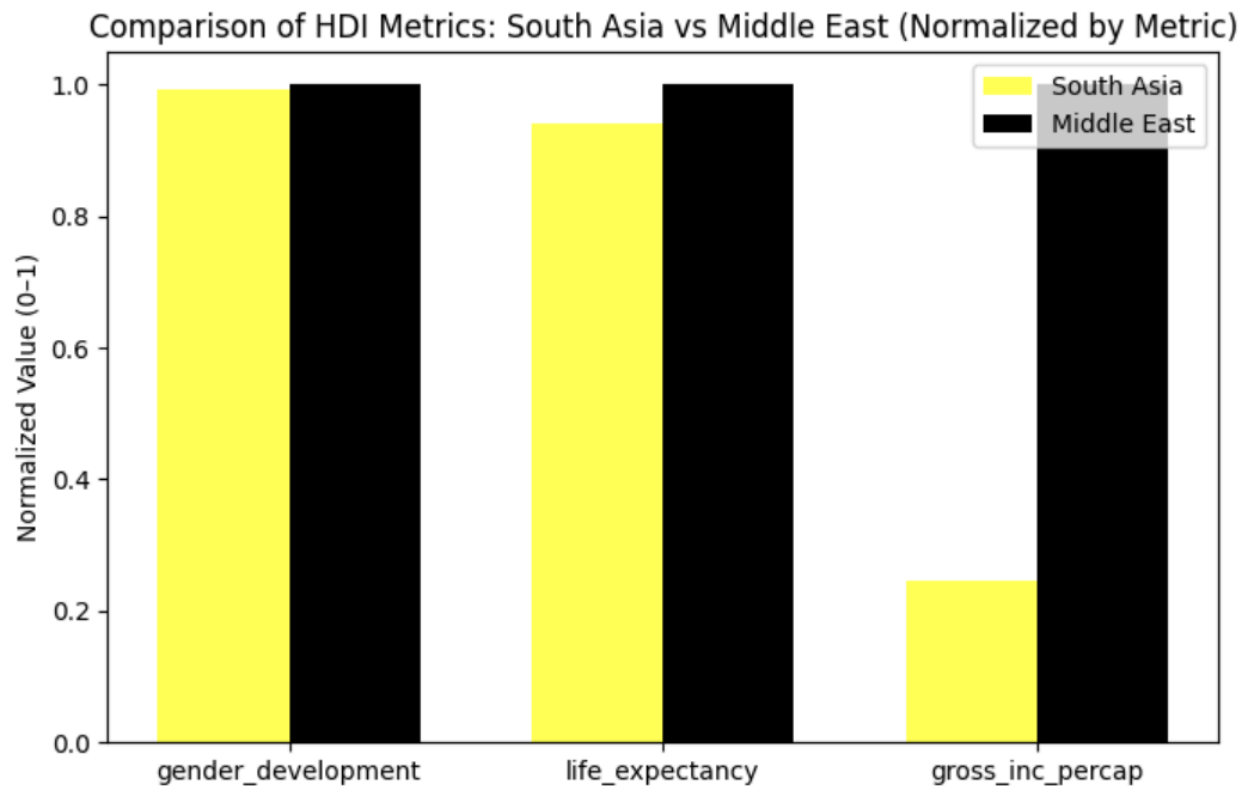


Fig11: Bar chart comparing HDI metrics between South Asia and Middle East

This bar chart compares the average values of three HDI metrics: gender development, life expectancy and gross income per capita between South Asia and Middle East. This visualization helps us identify the metric that have the largest development gaps between the regions.

Correlation Analysis

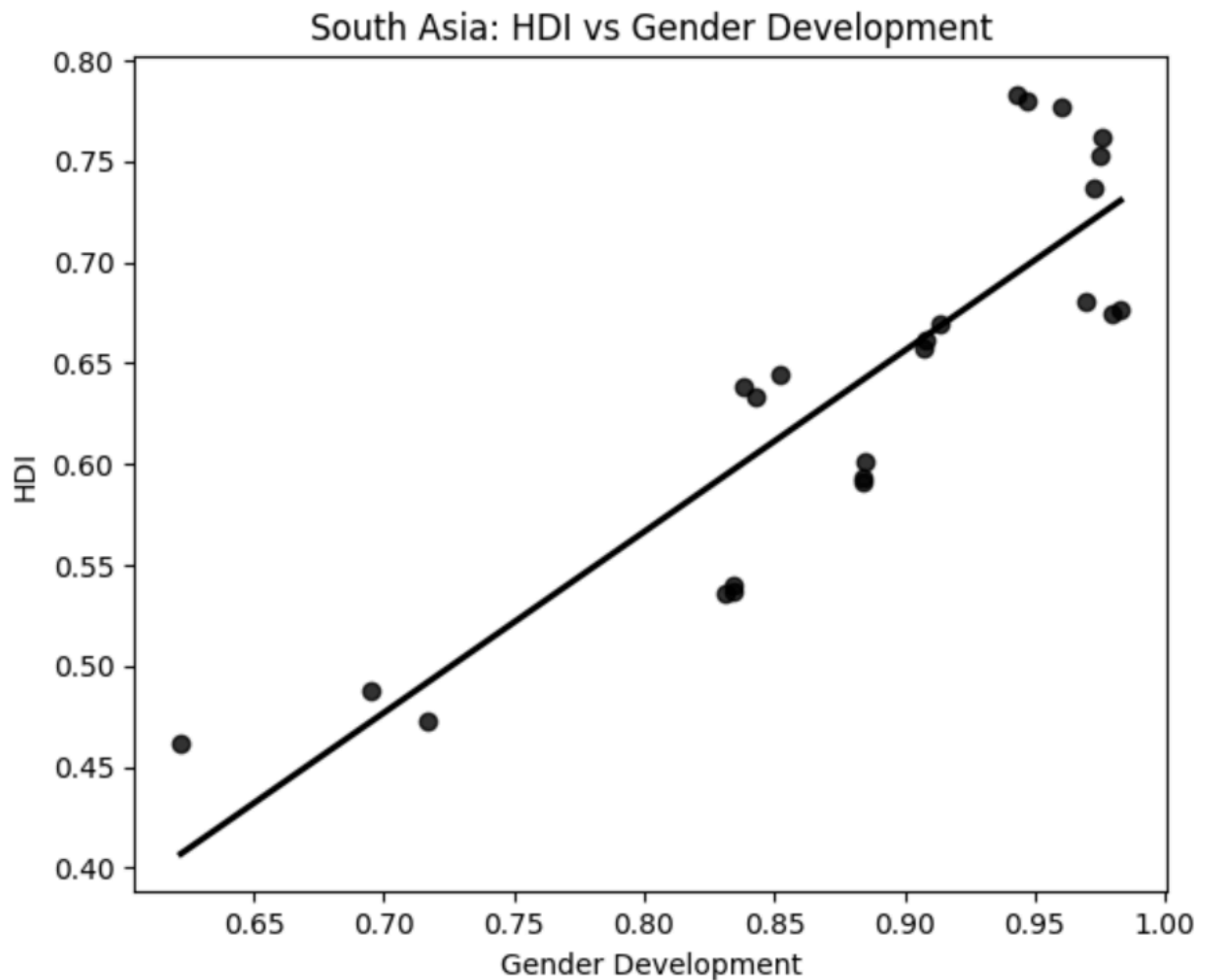


Fig13: Scatterplot showing the relationship between Gender Development and HDI in South Asia

This scatterplot shows how gender development relates to HDI in South Asian regions from 2020 – 2022. The regression line here indicates a positive relationship but the spread of data here shows that gender development alone does not fully impact the HDI in that region.

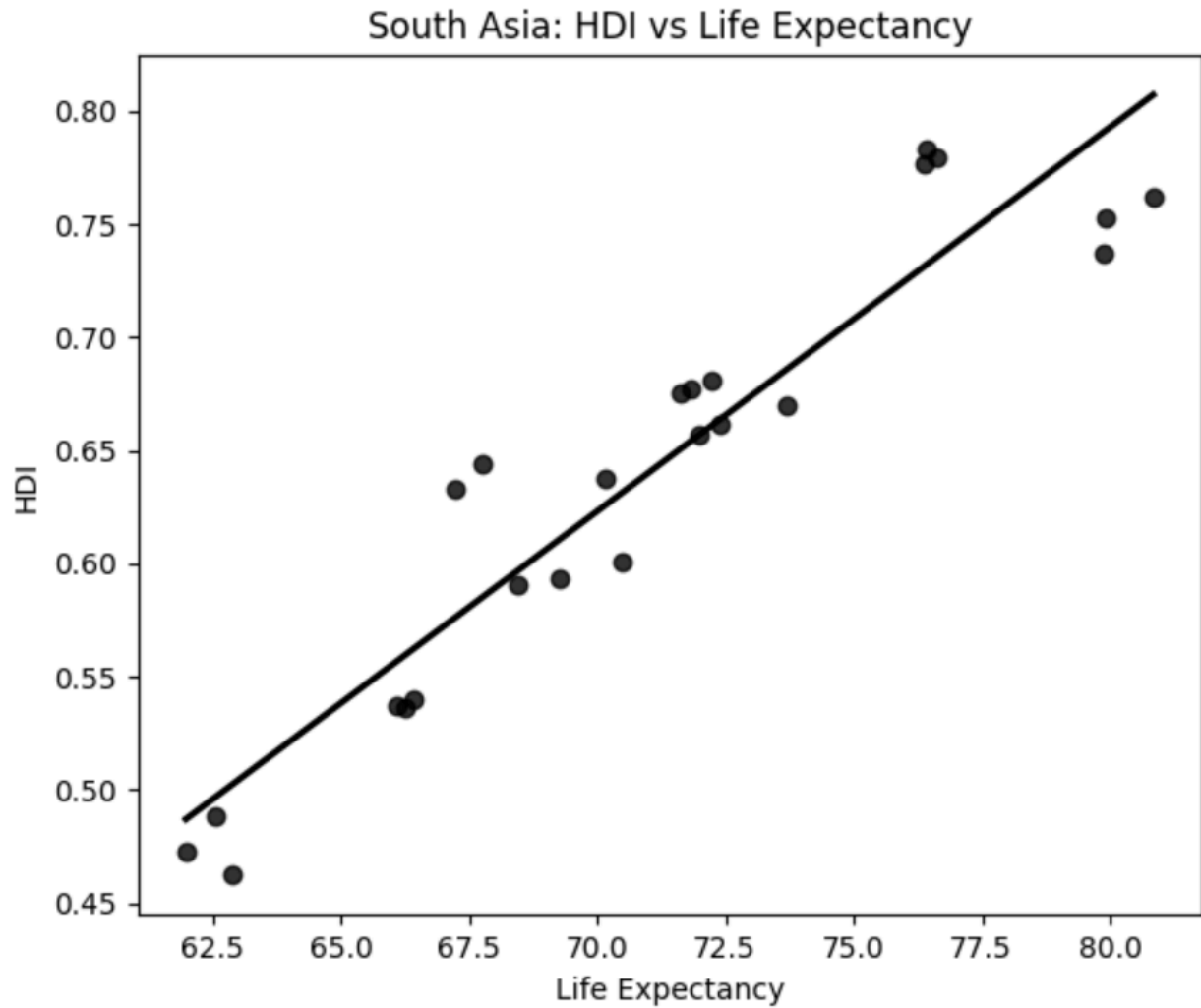


Fig14: Scatterplot showing the relationship between Life Expectancy and HDI in South Asia

This scatterplot shows a positive relationship between life expectancy and HDI in South Asia. This plot shows that health plays a crucial role in human development in the South Asian region.

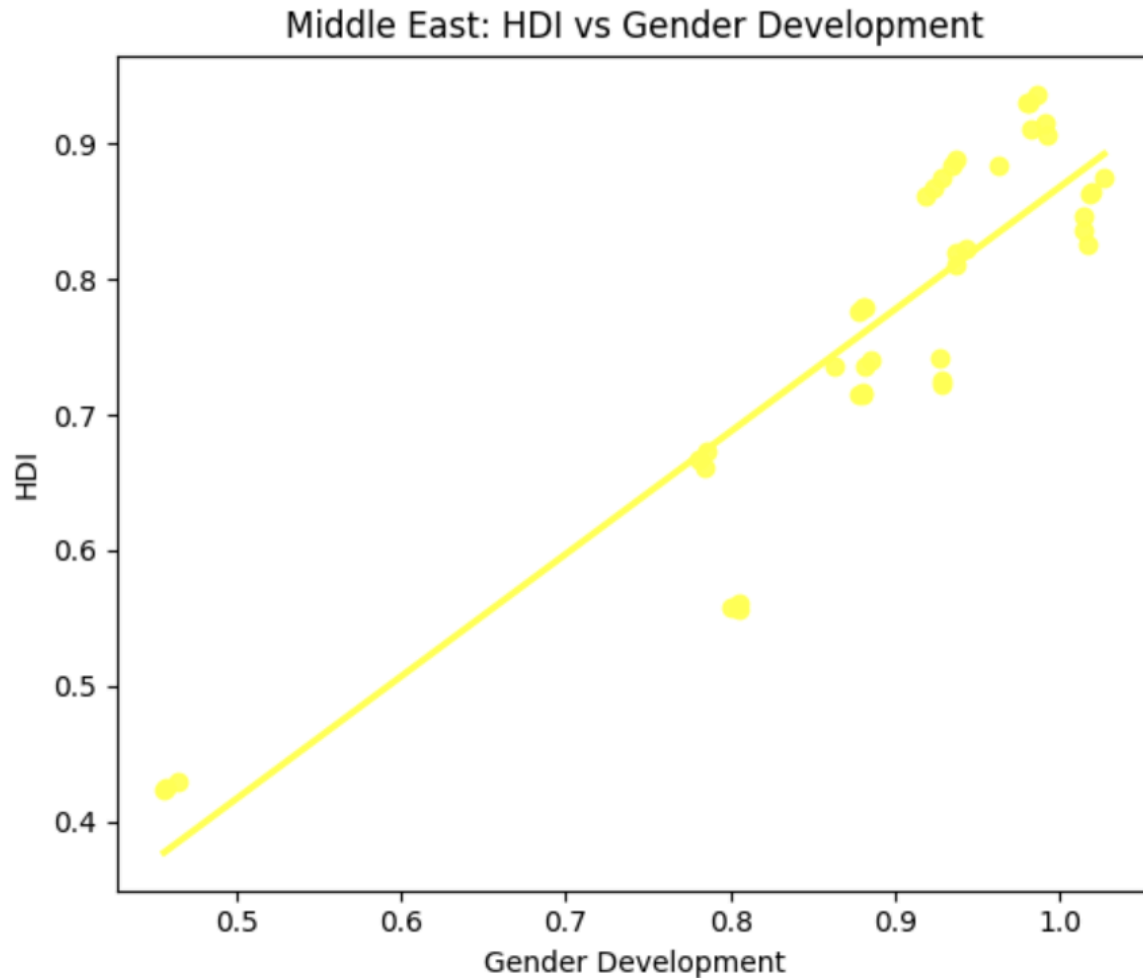


Fig15: Scatterplot showing the relationship between Gender Development and HDI in Middle East

This scatterplot shows the relationship between gender development and HDI for Middle East countries. This plot suggests a positive correlation meaning gender development does play a role in human development in this region.

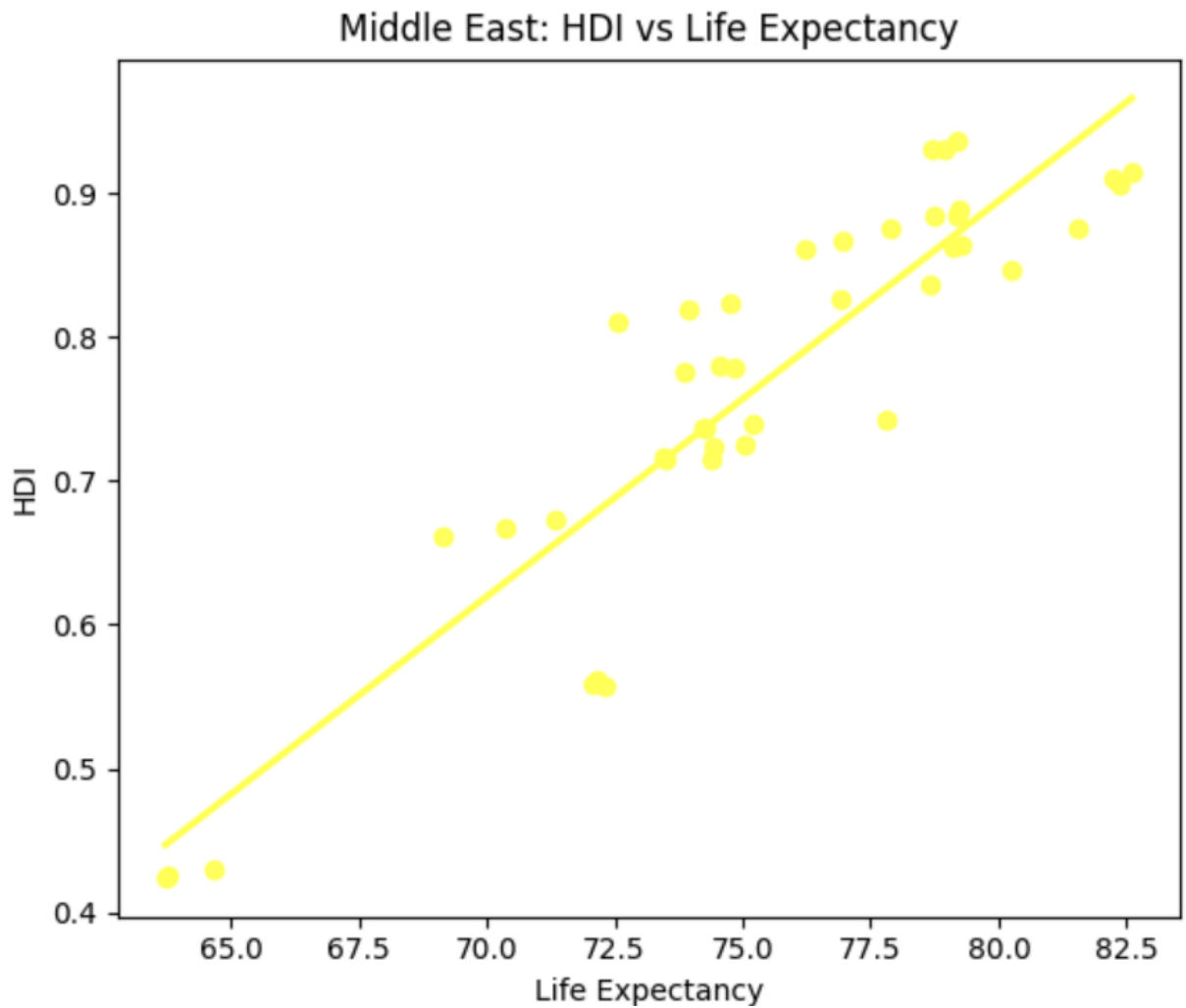


Fig16: Scatterplot showing the relationship between Life expectancy and HDI in Middle East

The plot highlights a strong relationship between life expectancy and HDI in the Middle East. This suggests that life expectancy is a key determinant of HDI in the region.

Outlier Detection:

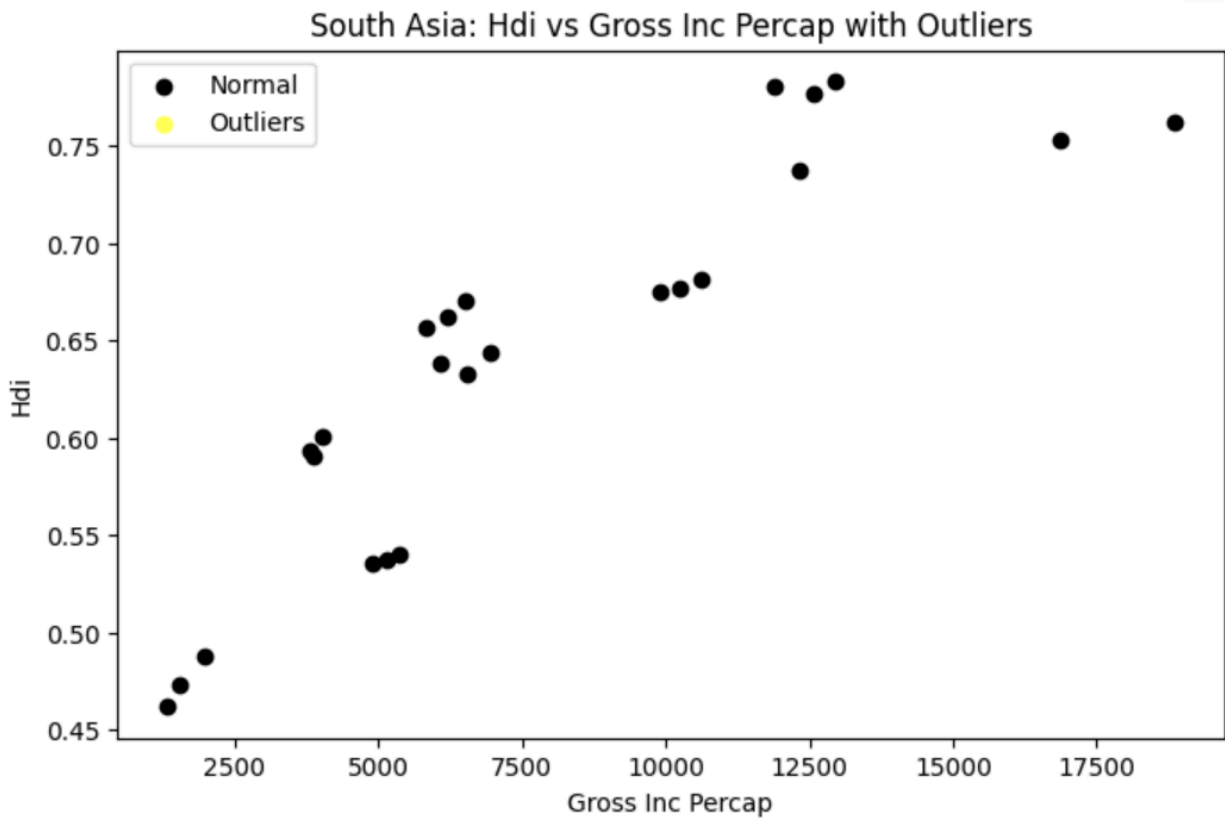


Fig17: Scatterplot identifying HDI and GNI per capita outliers in South Asian countries

This scatterplot shows the relationship between GNI per capita and HDI for South Asian countries, with data points colored to distinguish outliers from normal data using the IQR method. Since there are no outliers present, this shows a consistent development pattern in this region, with balanced HDI and GNI per capita values

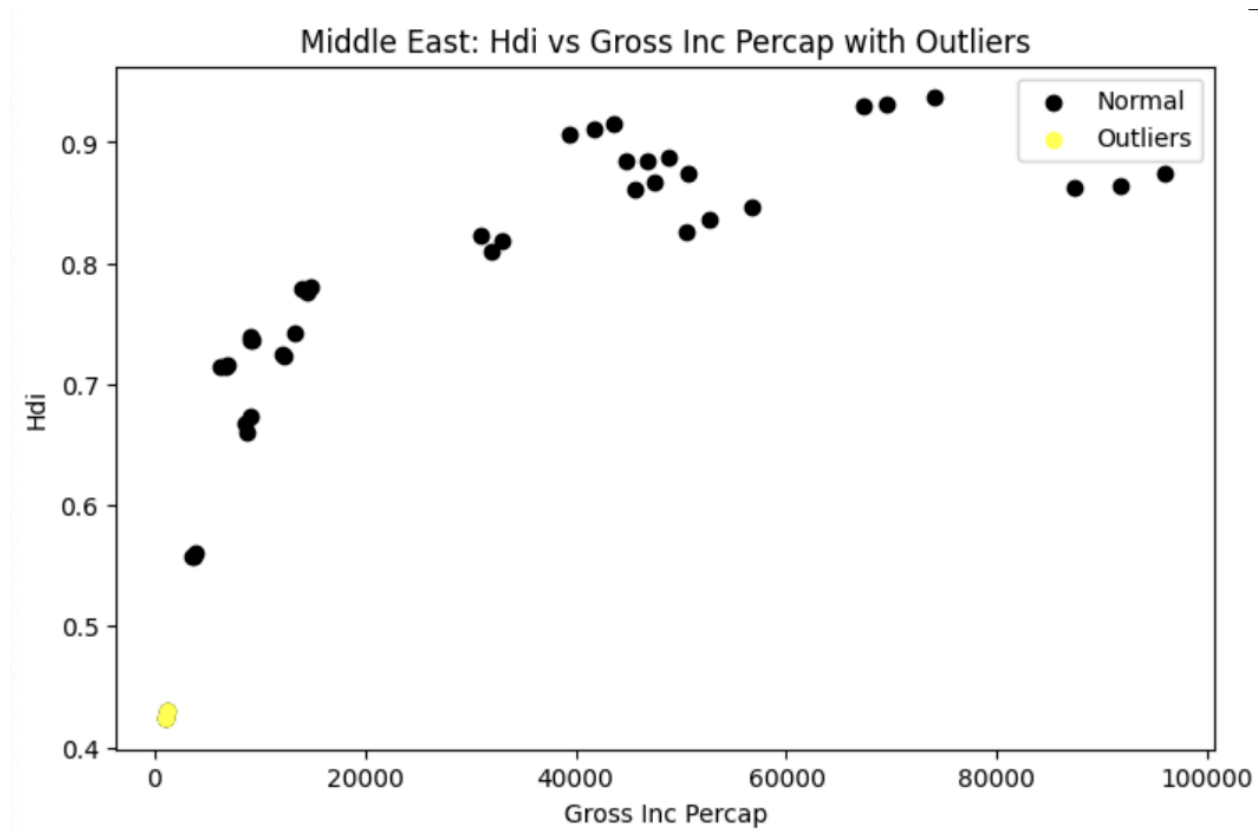


Fig18: Scatterplot identifying HDI and GNI per capita in Middle East countries

Interpretations and Discussions

This scatterplot shows the relationship between GNI per capita and HDI for Middle East, with data points colored to distinguish outliers from normal data using the IQR method. Compared to South Asia, a wider spread is visible in the data showing variability in income levels and also there is presence of outliers which suggest the presence of country with exceptionally low income relative to their HDI.

Descriptive Statistics

The descriptive analysis shows that the Middle East records a higher mean HDI than South Asia, indicating stronger overall human development in the Middle Eastern region. At the same time, the Middle East also displays a higher standard deviation, suggesting a wider spread of development outcomes among its countries. This implies that although the region performs better on average, development levels vary considerably within it. In contrast, South Asia demonstrates lower average HDI values but exhibits comparatively less variation, indicating more consistent development levels across its countries.

Top and Bottom Performers

The analysis of the highest- and lowest-performing countries reveals clear differences both within and between the two regions. Overall, Middle Eastern countries tend to achieve higher HDI scores than those in South Asia. However, when examining countries within each region, South Asia shows HDI values that are more closely clustered, suggesting a narrower development gap. Conversely, the Middle East exhibits a larger difference between its highest- and lowest-performing countries, highlighting greater inequality within the region.

Metric Comparisons

A comparison of key development indicators—gender development, life expectancy, and GNI per capita, further illustrates the contrasts between the two regions. Gender development levels are fairly similar, indicating minimal difference in this area. Life expectancy is generally higher in the Middle East, reflecting stronger healthcare outcomes. The most significant disparity is observed in GNI per capita, where the Middle East substantially exceeds South Asia. This indicates that income differences play a major role in shaping the overall development gap between the regions.

HDI Disparity

The assessment of HDI disparity indicates that the Middle East has a broader range of HDI values than South Asia, pointing to a larger gap between its highest- and lowest-performing countries. In comparison, South Asia shows a smaller HDI range, suggesting more evenly distributed development levels across the region. The higher coefficient of variation in the Middle East further confirms the presence of greater inequality in development outcomes.

Correlation Analysis

Correlation results reveal strong positive relationships between HDI and both gender development and life expectancy in South Asia and the Middle East. In South Asia, HDI is more strongly linked to life expectancy than to gender development, indicating that health-related factors play a particularly important role in influencing development outcomes. In contrast, the Middle East shows nearly equal correlations between HDI and both indicators, suggesting that health and gender development contribute similarly to human development in the region.

Outlier Detection

Outlier analysis highlights distinct development patterns between the two regions. In South Asia, no outliers were identified for either HDI or GNI per capita, indicating a stable and consistent relationship between income and human development. In the Middle East, although no income-related outliers were found, the presence of HDI outliers suggests uneven development levels among certain countries. This reflects greater variation in how development outcomes are achieved across the region.

Conclusion

This study analyzed global Human Development Index (HDI) trends from 1990 to 2022, with particular emphasis on the 2020–2022 period, to examine development patterns and regional inequalities. The results show that while many countries have attained high levels of human development, substantial disparities persist across regions. Developed nations demonstrated stable HDI trends, whereas developing regions especially South Asia experienced slower progress. The findings also confirm that although higher income generally supports improved HDI performance, economic growth alone does not determine human development. Social factors such as healthcare access, education, and gender equality play a critical role in shaping development outcomes. The comparison between South Asia and the Middle East further revealed that despite higher average HDI values in the Middle East, the region experiences greater internal inequality. Overall, the study emphasizes that long-term human development depends on balanced progress across multiple social and economic dimensions rather than reliance on income growth alone.

References

<https://hdr.undp.org/data-center/human-development-index#/indicies/HDI>

<https://hdr.undp.org/data-center/country-insights#/ranks>

APPENDIX:

https://github.com/prasannar2006/AI_assignment1_2550313.git

Introduction-1.pdf

Submission Details

Setting changes will not be saved.

UNIVERSITY OF WOLVERHAMPTON



Report On Prototype-1

Name: Prasanna Raj Kumar

Group: L5CG12

University Id: 2550313

Tutor: Raja Karki

Submitted on: 09/01/2026

Share





Page 1 of 31

27 Similarity Exclusions

19%

Overall Similarity

1 University of Wolverhampton on 2026-01-09 SUBMITTED WORKS 3%

2 University of Wolverhampton on 2026-01-07 SUBMITTED WORKS 3%

3 University of Wolverhampton on 2026-01-08 SUBMITTED WORKS 1%

4 University of Wolverhampton on 2026-01-08 SUBMITTED WORKS 1%

5 University of Wolverhampton on 2026-01-09 SUBMITTED WORKS <1%

6 University of Wolverhampton on 2026-01-09 SUBMITTED WORKS <1%

Setting changes will not be saved.

X



Report On Prototype-1

Name: Prasanna Raj Kumar
Group: LSCG12
University Id: 2550313
Tutor: Raju Karki
Submitted on: 09/01/2026

Share



Page 1 of 31

27 Similarity Exclusions

19%

Overall Similarity

1	University of Wolverhampton on 2026-01-09 SUBMITTED WORKS	3%
2	University of Wolverhampton on 2026-01-07 SUBMITTED WORKS	3%
3	University of Wolverhampton on 2026-01-08 SUBMITTED WORKS	1%
4	University of Wolverhampton on 2026-01-08 SUBMITTED WORKS	1%
5	University of Wolverhampton on 2026-01-09 SUBMITTED WORKS	<1%
6	University of Wolverhampton on 2026-01-09 SUBMITTED WORKS	<1%

>>

[Handwritten signature]